# Topic 1 - Fundamentals of the data-driven enterprise

This document is the handbook for Topic 1 – **Fundamentals of the data-driven enterprise** – within Module 1 – **Data Fundamentals**.

The purpose of this document is to guide your learning throughout this topic and help you to maximise the value you get from the materials provided by the BPP School of Technology.

## Context

This handbook is for one of 4 topics for this Module.


Every topic contributes towards the ultimate learning objectives for the Module, which you will be assessed on at the end of the term.

## Module Learning Outcomes

On successfully completing this module, you will be able to:

- **Recognise** fundamental data types and data source types, key Data Engineering technological standards and best practices, and relevant regulations
- **Apply** fundamental principles for fostering a data-driven culture through collaborating with diverse stakeholders in Agile and Lean teams, avoiding waste and respecting organisational processes for data quality and data project management

- **Model** standardised Big Data ecosystems and architectures for enterprise data using visual approaches and comprehend how to design rudimentary data products, and how they add value to the organisation
- **Explain** the importance of assessing, improving, and maintaining data quality using frameworks and methodologies to ensure data accuracy, completeness, and consistency.

# Module Assessment

The Level 5 Data Engineer EPA has two assessment methods, each with its own mapping of KSBs. The Assessment plan and assessment guidance documents above list the criteria and KSBs that are assessed. The criteria group the KSBs and describe what the apprentice needs to do to achieve a pass or distinction for that assessment method.

Both assessment methods need to be passed by the candidate:

### (1) Project with report

The learner will complete a project and write a report of 3500 words. Project brief submitted at gateway:

- Learners will have 10 weeks to complete the project and submit the report to the EPAO
- Learners also need to prepare and give a presentation to an independent assessor on their project
- The presentation with questions will last at least 50 minutes. The independent assessor will ask at least 6 questions about the project and presentation
- The project has to have real business application and benefit. Candidates are expected to showcase the use of appropriate standards for sustainability, privacy and security, thoroughly document their data pipeline designs, explain the choice of relevant tooling and demonstrate operational awareness of deployment, access control, risks, and how other stakeholders may be impacted positively and negatively

### (2) Professional discussion underpinned by a portfolio of evidence

- Learners will have a professional discussion with an independent assessor. It will last 80 minutes
- They will be asked at least 10 questions about Data Engineering
- The portfolio of evidence will be used to help answer the questions

- We expect the candidates to demonstrate examples of working with data teams on data projects and data products, showcase ideas for future-proofing data, be clear on applying problem-solving skills, show regulatory awareness, and sensitivity towards data quality, data governance and areas for continuous improvement, both personal and organisational

**Link to summative assessment:**

The "Fundamentals of the Data-Driven Enterprise" topic provides a strong foundation for the Summative Assessment by covering key concepts like data types, data management practices, standards, regulations, and best practices.

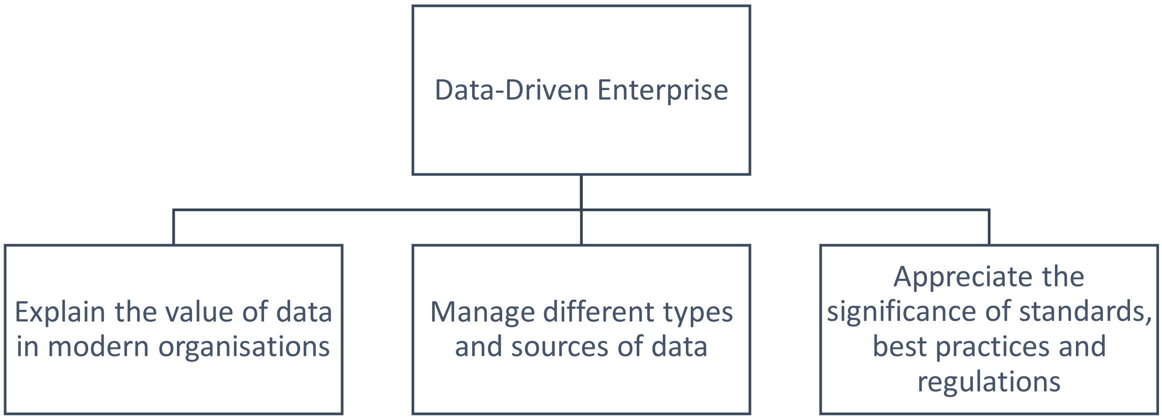
It aligns with the project requirements by explaining data pipeline design, documentation, and operational awareness. The professional discussion is supported through sections on data-driven culture, data quality, governance, and continuous improvement.

Specific areas of relevance include:

- Understanding qualitative and quantitative data types
- Managing data sources and the data lifecycle
- Adhering to technological standards (data formats, APIs, cloud computing)
- Complying with regulations (GDPR, HIPAA, ISO 27001)
- Applying data stewardship principles (quality, governance, ethics)
- Leveraging engineering best practices (scalability, reliability, security)
- Integrating diverse data sources and visualising insights

# Topic Learning Outcomes

As a step towards build your skills towards the final module assessment, the learning objectives for this topic are:




# Introduction

In the modern, data-driven business landscape, mastering data fundamentals is pivotal for organisations to harness the true power of their data assets. This is especially crucial for data engineers, whose roles are centred around designing, building, and maintaining the robust data infrastructures that form the bedrock of data-driven decision-making.

By gaining a comprehensive grasp of key data engineering concepts covered in this topic - from data types, formats and measurement units, to data collection, cleansing, transformation and integration techniques - data engineers are empowered to navigate the ever-increasing complexities of managing data at scale.

Moreover, adhering to relevant industry standards, regulations and governance frameworks is paramount to ensuring compliance and fostering trust among stakeholders. An in-depth understanding of data engineering principles like scalability, reliability and security enables practitioners to architect systems capable of handling immense data volumes while maintaining performance and resilience. Throughout this topic, case studies illuminate how organisations are leveraging these fundamentals to drive tangible business impact.

A prime example is the media streaming giant Netflix, which has established itself as a pioneer in harnessing data to optimise decision-making across its operations. By meticulously collecting, analysing and drawing insights from vast troves of subscriber data – spanning viewing habits, preferences and feedback – Netflix's data engineers have equipped the company to make informed decisions on content licensing, recommendation engines and product enhancements, thereby delivering unparalleled customer experiences.

Upon completing this topic, you will possess a comprehensive skillset encompassing data visualisation tools, techniques for extracting business value from data, and strategies for fostering a data-driven organisational culture. Armed with this robust foundation, you will be poised to contribute effectively to transformative, data-centric projects, collaborate seamlessly with cross-functional teams, and spearhead groundbreaking data-driven initiatives that yield sustainable competitive advantages for organisations.

# Structure

Topics for this programme follow a Prepare-Collaborate-Apply structure:

# Prepare

This is the stage where you build the knowledge to underpin your learning. This might involve completing interactive e-learning packages, watching videos, or working through reading materials.

It is essential that you make the most of the learning materials provided before attending webinars, as this will allow you to test your knowledge and stretch you understanding further.

The e-learning for this topic covers building a data-driven culture, fundamentals of data types and sizes, standards and best practices in data engineering, working with different data sources and types through the data management lifecycle, and utilisation of visualisation tools.

# Collaborate

This is where you will receive guidance from our expert tutors and coaches to shape and refine your understanding through in-depth explanation, discussion, testing and carrying out more advanced practical and realistic tasks. This also helps to develop valuable team-working skills.

The webinar for this topic will focus on building a data-driven culture, data fundamentals like types/sizes, key standards and best practices in data engineering, a practical lab, and summarised the core learning concepts.

# Apply

You now apply the knowledge you have developed to real-world tasks.

## Off-the-job learning tasks

This stage is all about ensuring you truly grasp and retain what you've learned. Through completion of off-the-job (OTJ) revision tasks and tests, you'll get plenty of practice applying your knowledge. Plan to dedicate 6-8 hours each week to guided study and portfolio work, with sessions typically on the same day each week.

### **Task 1 brief:** Creating your learning journal in GitHub

This task is about creating a learning journal on GitHub to document your learning journey throughout the course. It is divided into the following stages:

- **Stage A:** Creating a GitHub account and explore the platform
- **Stage B:** Setting up a directory structure for your repository by creating folders for each module
- **Stage C:** Creating a markdown file within the "Data Fundamentals" folder to compile your notes from the first webinar. Use proper formatting, headings, lists, and visual aids
- **Stage D:** Structure your learning from the webinar by listing key objectives, capturing main concepts, noting questions, self-researching answers, and reflecting on real-world applications

The task focuses on the benefits of keeping a learning journal, such as deepening understanding, boosting memory, tracking progress, and identifying knowledge gaps. It also provides tips for success, like committing changes regularly, seeking feedback, and keeping notes organised.

### Further guidance can be found here: [L5DE 1.1 Apply (OTJ)]

### **Task 2 brief:** Data types and data sources used within your organisation

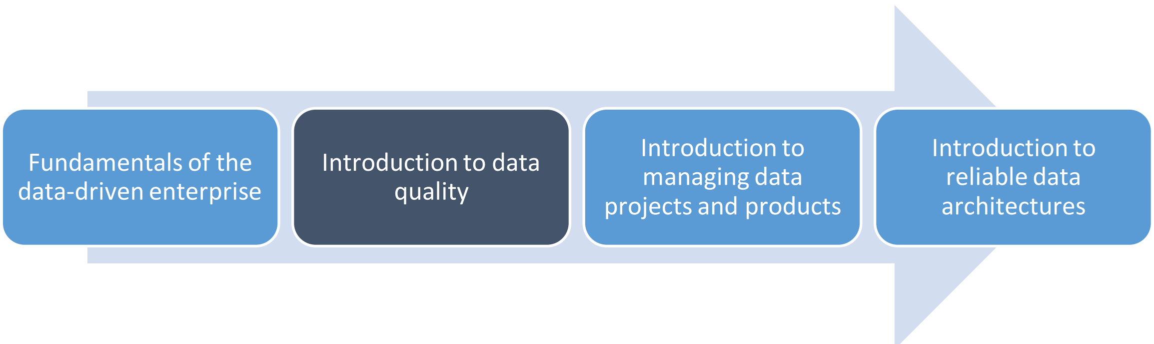
For this task you should have at least one meeting with an experienced Data Engineer (or Software Engineer) to discuss and identify the key data types and data sources used within your organisation's data systems.

Following this meeting, you should:

- Outline (e.g. in an email) an individual approach for learning the standards, best practices and regulations which inform the use and management of data systems within the organisation
- Cross-validate your findings with other Data Engineers (e.g., in a meeting, or email). Document your learning in your learning journal

# Link

This handbook is for one of 4 topics for this Module.




The sequence of topics in this module is carefully designed so that your knowledge and skills will develop as you progress.

The next topic is **Introduction to data quality**.